



Lecture_6

Windows Programming_II

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Introduction to ADO.NET

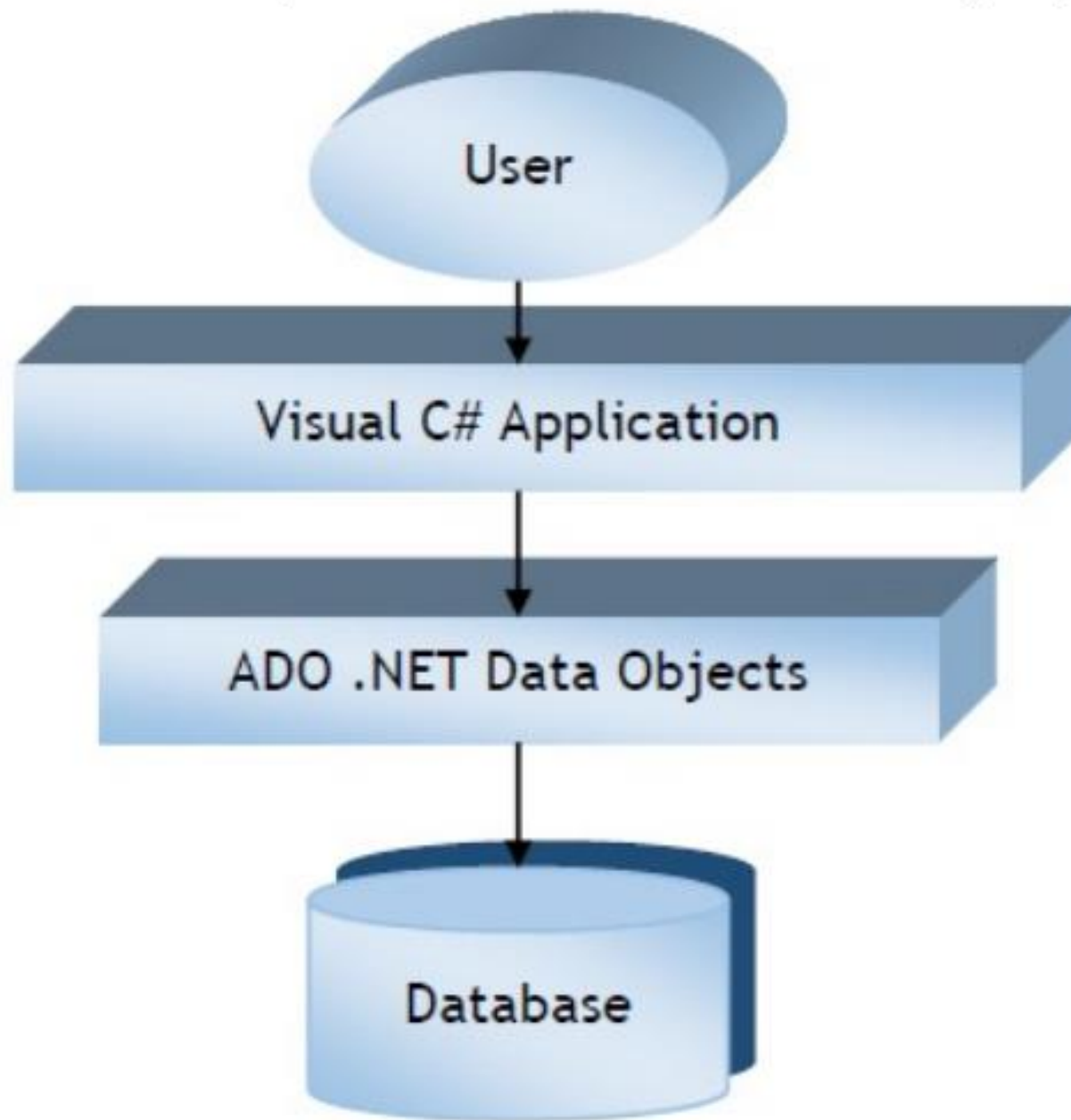


Use a SQL Server database in a Windows application

- ☐ Form application and database connection
 - Get data from a database and display it on Windows Forms
 - Insert data Into a database table
 - Delete data from a database table
 - Update database rows by windows forms

Introduction

- For database management, we say our Visual C# application acts as a front-end to the database. This means the Visual C# application provides the interface between the user and the database. This interface allows the user to tell the database what he or she needs and allows the database to respond to the request displaying the requested information in some manner.
- A Visual C# application cannot directly interact with a database. There is a set of intermediate components between the application and the database known as ADO (ActiveX Data Object) .NET data objects

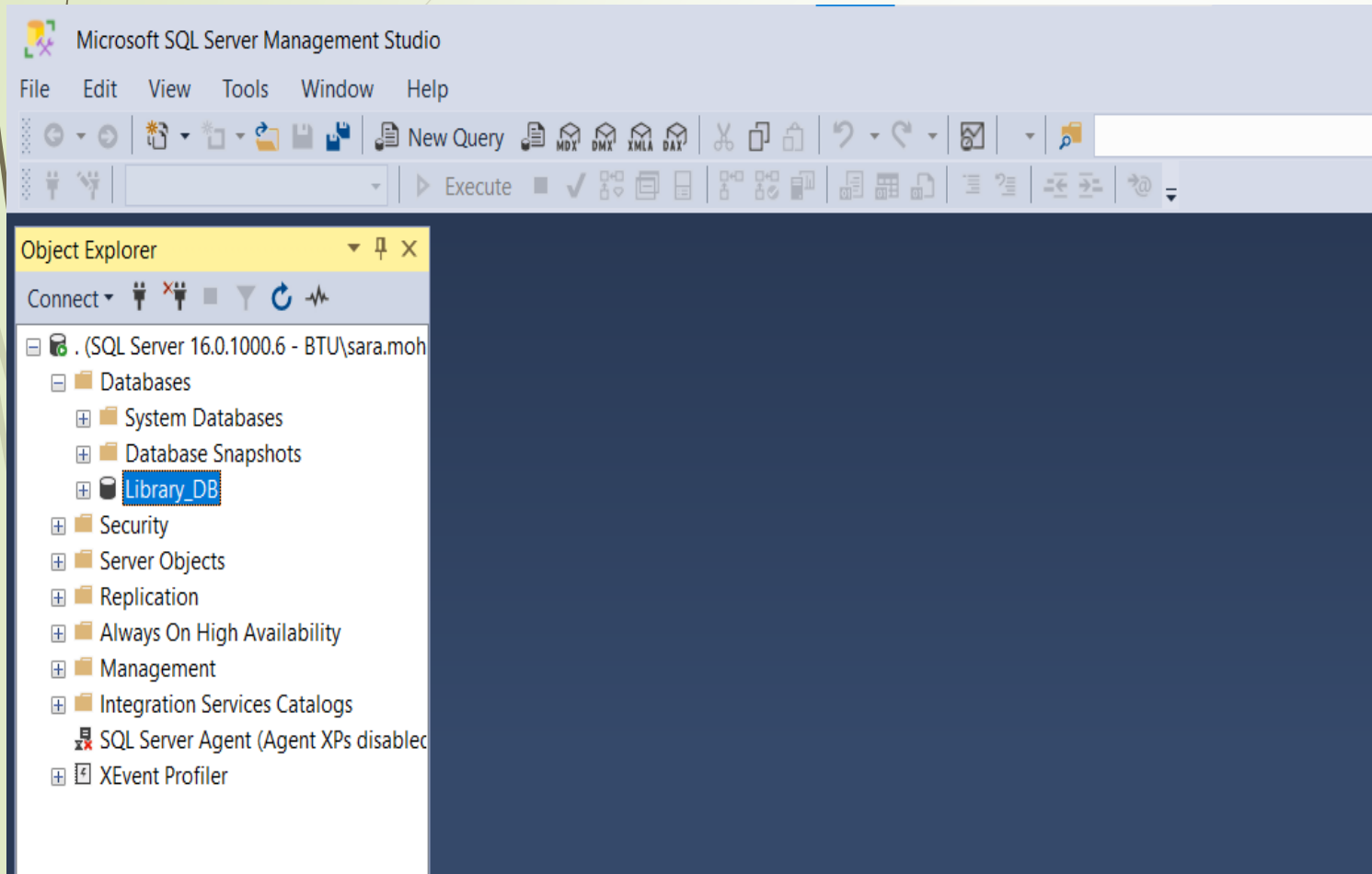




The **data objects** are Visual C# components that allow connection to the database, creation of data sets from the database and management of the database contents. These objects are the conduit between the application and the database, passing information back and forth between the two.

- There are many commercial products (Access, SQL Server, Oracle) that do database management tasks.
- **Connection objects and command objects** are specific to the type of database being used. Visual C# provides objects that work with Object Linking and Embedding Databases (OLE DB), SQL Server, including the Microsoft Data Engine (MSDE), Open Database Connectivity (ODBC) and Oracle databases.
- Here you need to download **SQL server and a database management system(Sql Management studio)**

Sql Management studio will be used to create and manage SQL databases,



Each database requires some way for a user to interact with the information within.

Such interaction is performed by a **database management system** (DBMS).

Connection Object : The connection object establishes a connection between your application and the database.



As mentioned, Visual C# supports several databases types. Hence, it offers several different connection objects: OleDbConnection, SqlConnection, OdbcConnection, OracleConnection.



Connection object **properties**:

ConnectionString	Contains the information used to establish a connection to a database.
Database	Name of the current database.
DataSource	Name of the current database file.
State	Current connection state (Closed, Connecting, Open, Executing, Fetching, Broken)

Connection object **methods**:

Open	Open the connection.
Close	Close the connection.
Dispose	Dispose of the connection object.



System.Data.SqlClient namespace

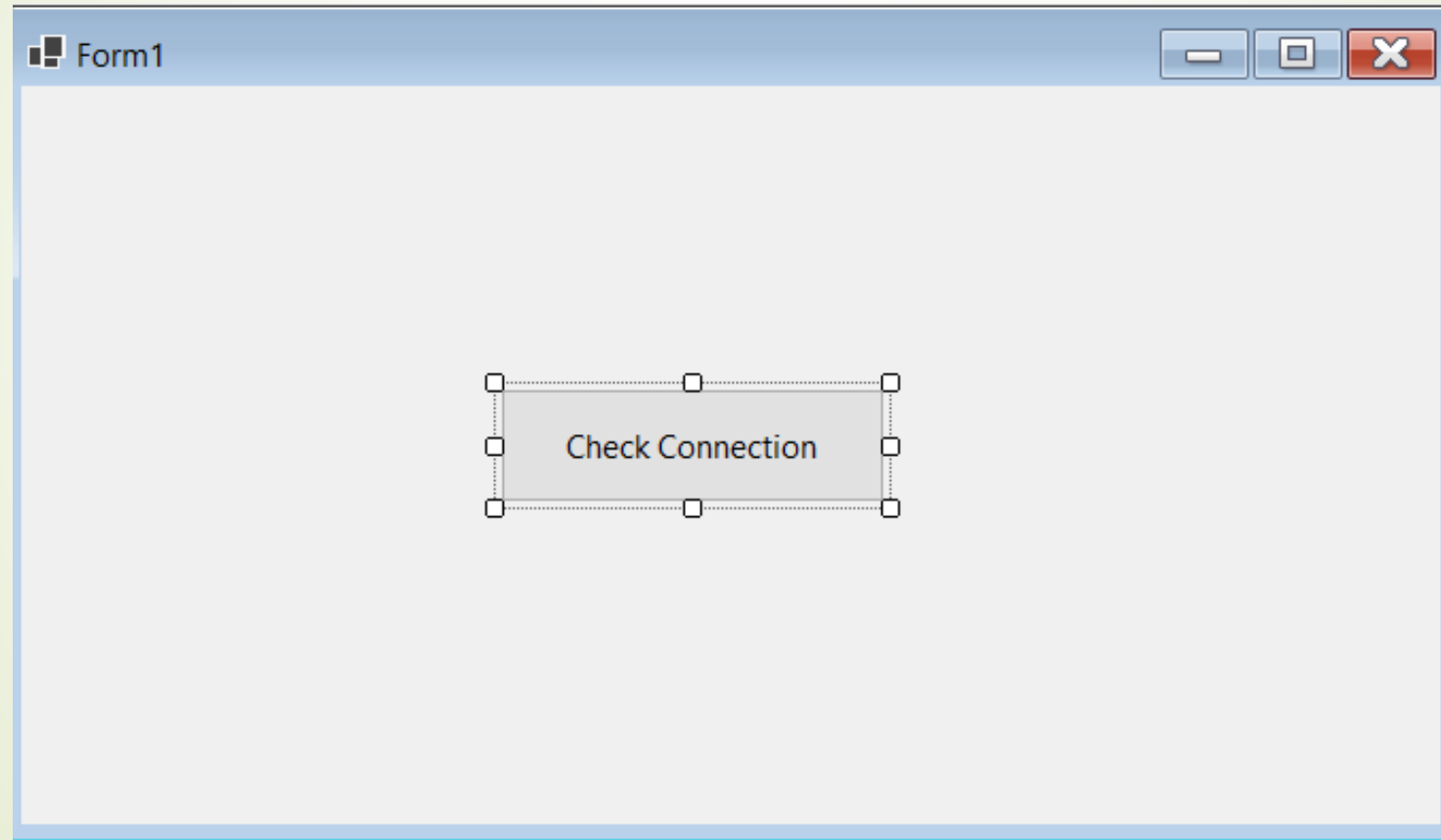
- This interacts with SQL Server. It allows the development of data-driven applications.
 - The types we use include **SqlConnection**, **SqlCommand**, and **SqlDataReader**.
- 

SqlConnection Class

- Represents a connection to a SQL Server database. This class cannot be inherited.
- It is found in `System.Data.SqlClient` namespace.
- This class has the following constructors:

<code>SqlConnection()</code>	Initializes a new instance of the <code>SqlConnection</code> class.
<code>SqlConnection(String)</code>	Initializes a new instance of the <code>SqlConnection</code> class when given a string that contains the connection string.
<code>SqlConnection(String, SqlConnectionCredential)</code>	Initializes a new instance of the <code>SqlConnection</code> class given a connection string, that does not use <code>Integrated Security = true</code> and a <code>SqlConnectionCredential</code> object that contains the user ID and password.

Server and Database connection to the window form example



Form1.cs [Design]*

connection

DataBase_Connection.Form1

conn

```
using System.Data.SqlClient;

namespace DataBase_Connection
{
    3 references
    public partial class Form1 : Form
    {
        SqlConnection conn = new SqlConnection("Data Source=l1-ict;Initial Catalog=Library_DB;Integrated Security=true");
        1 reference
        public Form1()
        {
            InitializeComponent();
        }

        1 reference
        private void Form1_Load(object sender, EventArgs e)
        {
        }

        1 reference
        private void button1_Click(object sender, EventArgs e)
        {
            conn.Open();
            MessageBox.Show("the connection state: " + conn.State);
            conn.Close();
        }
    }
}
```

You should download the System.Data.SqlClient to your application to handle database (SQL Server)

Connection statement by create SqlConnection object which takes:

- Server name
- Database name
- Integrated security : describe the way you enter to server if it protected by user and password the value will be false

Open() method to open connection with the database
State will get the current state for the connection
Close() method to close connection

Create the following table in Library_DB

The screenshot displays the SQL Server Enterprise Manager interface. On the left, the 'Object Explorer' pane shows the hierarchy: 'Databases' > 'Library_DB' > 'Tables' > 'dbo.Books_TB'. The 'dbo.Books_TB' table is selected. The main pane on the right shows the table's structure with the following columns:

Column Name	Data Type	Allow Nulls
ID	nvarchar(50)	☐
Book_Title	nvarchar(250)	☐
Author	nchar(30)	☐
Publish_date	datetime	☒
Page_number	int	☒
		☐

Fill the table like this

100 %



Results



Messages

	ID	Book_Title	Author	Publish_date	Page_number
1	1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
2	2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
3	3	History of Data	Break Warson	2021-05-03 00:00:00.000	100

SQLQuery2.sql - (I...\sara.mohamed (55))*

```
select * from Books_TB
```

100 %

Results Messages

	ID	Book_Title	Author	Publish_date	Page_number
1	1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
2	2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
3	3	History of Data	Break Warson	2021-05-03 00:00:00.000	100

SQLQuery2.sql - (I...\sara.mohamed (55))*

```
select Book_Title ,Author from Books_TB where ID='2'
```

100 %

Results Messages

	Book_Title	Author
1	Programing in Java	Tomas Bear

SQLQuery2.sql - (I...\sara.mohamed (55))*

```
select Book_Title from Books_TB
```

100 %

Results Messages

	Book_Title
1	C++ basics
2	Programing in Java
3	History of Data

SQLQuery2.sql - (I...\sara.mohamed (55))*

```
select Book_Title ,Author from Books_TB
```

100 %

Results Messages

	Book_Title	Author
1	C++ basics	Jine Pual
2	Programing in Java	Tomas Bear
3	History of Data	Break Warson

```
select Book_Title ,Author from Books_TB order by Author
```

100 %

Results Messages

	Book_Title	Author
1	History of Data	Break Warson
2	C++ basics	Jine Pual
3	Programing in Java	Tomas Bear

```
select Book_Title ,Author from Books_TB order by Author desc
```

100 %

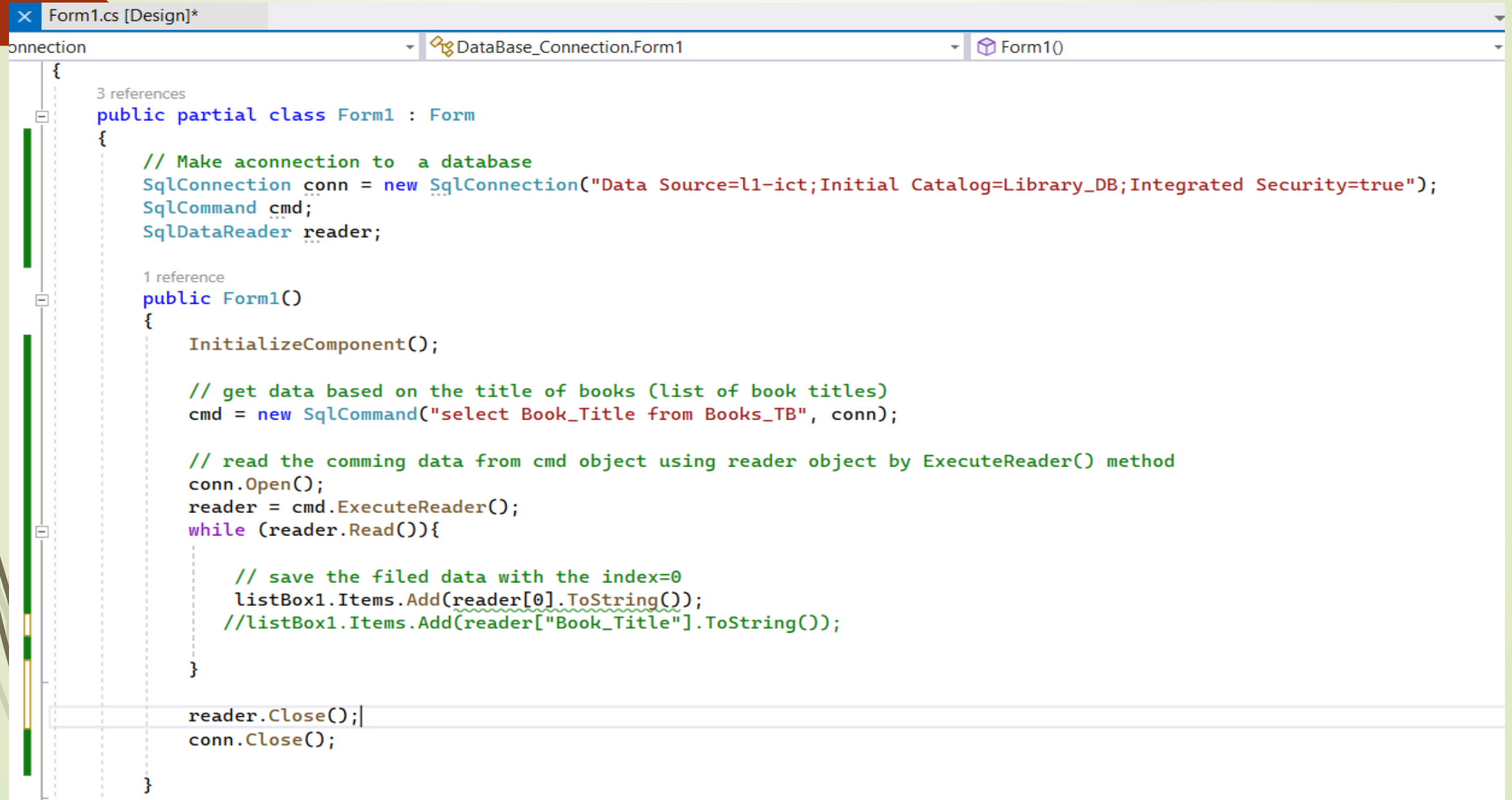
Results Messages

	Book_Title	Author
1	Programing in Java	Tomas Bear
2	C++ basics	Jine Pual
3	History of Data	Break Warson

SqlCommand class

- ➡ The command object is used to define commands to send to the database. In this course, these commands will be SQL queries.
- ➡ To insert data to a database table
- ➡ To get data from database
- ➡ Delete the data from the data table

Example for using SqlCommand class



```
Form1.cs [Design]*
    DataBase_Connection.Form1
    Form1()

{
    3 references
    public partial class Form1 : Form
    {
        // Make aconnection to a database
        SqlConnection conn = new SqlConnection("Data Source=ll-ict;Initial Catalog=Library_DB;Integrated Security=true");
        SqlCommand cmd;
        SqlDataReader reader;

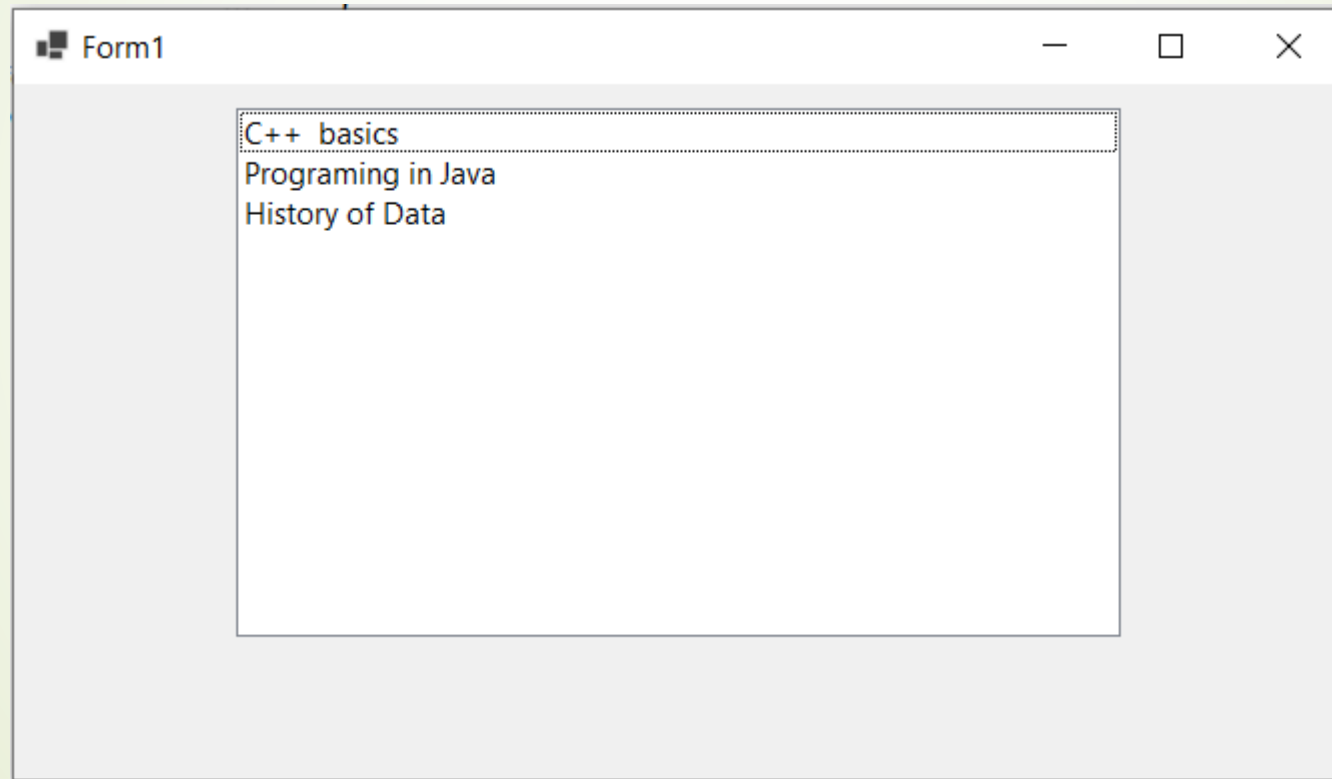
        1 reference
        public Form1()
        {
            InitializeComponent();

            // get data based on the title of books (list of book titles)
            cmd = new SqlCommand("select Book_Title from Books_TB", conn);

            // read the comming data from cmd object using reader object by ExecuteReader() method
            conn.Open();
            reader = cmd.ExecuteReader();
            while (reader.Read()){
                // save the filed data with the index=0
                listBox1.Items.Add(reader[0].ToString());
                //listBox1.Items.Add(reader["Book_Title"].ToString());
            }

            reader.Close();
            conn.Close();
        }
    }
}
```

Example: read the book titles and display on ListBox



```
SqlDataReader reader;
```

1 reference

```
public Form1()
```

```
{
```

```
    InitializeComponent();
```

```
    // get data based on the title of books (list of book titles)
```

```
    cmd = new SqlCommand("select Book_Title from Books_TB", conn);
```

```
    // read the coming data from cmd object using reader object by ExecuteReader() method
```

```
    conn.Open();
```

```
    reader = cmd.ExecuteReader();
```

```
    while (reader.Read()){
```

```
        // save the      data with the index=0
```

```
        listBox1.Items.Add(reader.GetString(0));
```

```
    }
```

```
    reader.Close();
```

```
    conn.Close();
```

```
}
```

1 reference

```
private void Form1_Load(object sender, EventArgs e)
```

```
{
```

```
}
```

Form1

C++ basics

Programing in Java

History of Data

Get multiple values from data table

```
Form1.cs [Design]
- DataBase_Connection.Form1
- Form1()

3 references
public partial class Form1 : Form
{
    // Make aconnection to a database
    SqlConnection conn = new SqlConnection("Data Source=ll-ict;Initial Catalog=Library_DB;Integrated Security=true");
    SqlCommand cmd;
    SqlDataReader reader;

    1 reference
    public Form1()
    {
        InitializeComponent();

        // get data based on the title of books (list of book titles)
        cmd = new SqlCommand("select * from Books_TB", conn);

        // read the comming data from cmd object using reader object by ExecuteReader() method
        conn.Open();
        reader = cmd.ExecuteReader();
        while (reader.Read()){
            --
            listBox1.Items.Add(reader.GetString(1)+"====|"+reader.GetString(2)+" "+reader.GetSqlDateTime(3));
        }

        reader.Close();
        conn.Close();
    }
}
```

```
SqlCommand cmd;  
SqlDataReader reader;
```

1 reference

```
public Fo
```

 class System.Data.SqlClient.SqlDataReader

Provides a way of reading a forward-only stream of rows from a SQL Server database. This class cannot be inherited.



insert rows into a data table

SQLQuery3.sql - (I...\sara.mohamed (60))* X SQLQuery2.sql - (I...\sara.mohamed (55))*

```
insert into Books_TB(ID,Book_Title,Author,Publish_date,Page_number)
values (4,'Python','johanson mark',30/10/2019,600)
```

100 %

Results Messages

	ID	Book_Title	Author	Publish_date	Page_number
1	1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
2	2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
3	3	History of Data	Break Warson	2021-05-03 00:00:00.000	100
4	4	Python	johanson mark	1900-01-01 00:00:00.000	600

Type the book data by this form then the row data will be saved in the data table

The image shows a Windows application window titled "Form1". Inside the window is a form titled "Add Book". The form contains the following fields and controls:

- Book_ID**: A text input field.
- Book_Title**: A text input field.
- Author**: A text input field.
- Number of Pages**: A text input field.
- Data Of Puplising**: A date picker control showing "04 April 2024".
- Buttons**: "Cancel" and "Add" buttons at the bottom of the form.

InsertData.Form1

btn_add_Click(object sender, EventArgs e)

```
namespace InsertData
```

```
{
```

3 references

```
public partial class Form1 : Form
```

```
{
```

```
    // Make aconnection to a database
```

```
    SqlConnection conn = new SqlConnection("Data Source=l1-ict;Initial Catalog=Library_DB;Integrated Security=true");
```

```
    SqlCommand cmd;
```

```
    SqlDataReader reader;
```

1 reference

```
    public Form1()
```

```
    {
```

```
        InitializeComponent();
```

```
    }
```

1 reference

```
    private void btn_cancel_Click(object sender, EventArgs e)
```

```
    {
```

```
        this.Close();
```

```
    }
```

1 reference

```
private void btn_add_Click(object sender, EventArgs e)
{
    try
    {
        cmd = new SqlCommand("insert into Books_TB(ID, Book_Title, Author, Publish_date, Page_number)values" +
            " (' " + txt_id.Text + "',' " + txt_title.Text + "',' " + txt_author.Text + "',' " + dateTimePicker1.Value + "',' " + txt_num.Text+ ")", conn);
        conn.Open();
        cmd.ExecuteNonQuery();
        MessageBox.Show("the row has been added to the database table");

    }catch(SqlException ms)
    {
        MessageBox.Show("there is Error:"+ms.Message);
    }
    finally
    {
        conn.Close();
    }
}
```


Form1

Add Book

Book_ID

6

Book_Title

c++ programming

Author

ali morad

Number of Pages

345

Data Of Puplising

01 May 2022



Cancel

Add

100 %

Results

Messages

	ID	Book Title	Author	Publish date	Page number
1	6	c++ programming	ali morad	2022-01-05 21:30:17.000	345
2	1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
3	2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
4	3	History of Data	Break Warson	2021-05-03 00:00:00.000	100
5	4	Python	johanson mark	1900-01-01 00:00:00.000	600
6	5	Python	johanson mark	1900-01-01 00:00:00.000	600



Delete rows from a data table

SQLQuery3.sql - (I...\sara.mohamed (60))*

SQLQuery2.sql - (I...\sara.mohamed (55))*

```
Delete from Books_TB where ID=5  
select * from Books_TB
```

100 %

Results

Messages

	ID	Book_Title	Author	Publish_date	Page_number
1	6	c++ programming	ali morad	2022-01-05 21:30:17.000	345
2	1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
3	2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
4	3	History of Data	Break Warson	2021-05-03 00:00:00.000	100
5	4	Python	johanson mark	1900-01-01 00:00:00.000	600



Delete the row of id=3

	ID	Book_Title	Author	Publish_date	Page_number
1	6	c++ programming	ali morad	2022-01-05 21:30:17.000	345
2	1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
3	2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
4	3	History of Data	Break Warson	2021-05-03 00:00:00.000	100
5	4	Python	johanson mark	1900-01-01 00:00:00.000	600

1 reference

```
private void button1_Click(object sender, EventArgs e)
{
    try
    {
        cmd = new SqlCommand("Delete From Books_TB where ID='" + txt_Id.Text + "'", conn);
        conn.Open();
        cmd.ExecuteNonQuery();
        MessageBox.Show("Row is deleted from Database Table");
    } catch (SqlException ex)
    {
        MessageBox.Show(ex.Message);
    }
    finally
    {
        conn.Close();
    }
}
```

Form1

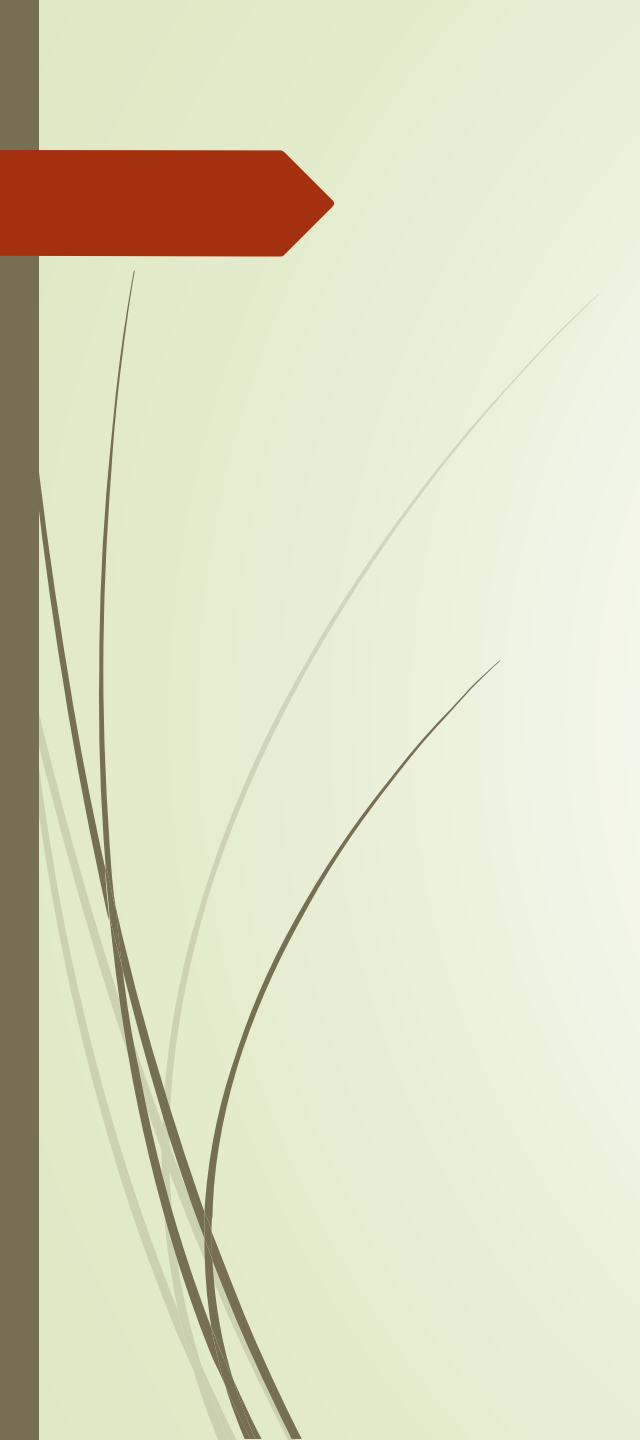
Enter ID Nunner:

Row is deleted from Database Table

100 %

Results Messages

	ID	Book_Title	Author	Publish_date	Page_number
1	6	c++ programming	ali morad	2022-01-05 21:30:17.000	345
2	1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
3	2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
4	4	Python	johanson mark	1900-01-01 00:00:00.000	600



Update the data table rows

Update example

Form1

Enter the ID of Book:

Book Details

ID

Book_Title

Author

Published Date

Page Numbers

1 reference

```
private void btn_show_Click(object sender, EventArgs e)
{
    try
    {
        cmd = new SqlCommand("Select * from Books_TB where ID='" + txtshow.Text + "'", conn);
        conn.Open();
        reader = cmd.ExecuteReader();
        reader.Read();
        texttitle.Text = reader["Book_Title"].ToString();
        textauthor.Text = reader["Author"].ToString();
        dateTimePicker1.Text = reader["Publish_date"].ToString();
        textnumbers.Text = reader["Page_number"].ToString();

    } catch (SqlException ex)
    {
        MessageBox.Show(ex.Message);
    }
    finally
    {
        reader.Close();
        conn.Close();
    }
}
```

When type the id of book , the data of this number will be shown on the form

Form1

Enter the ID of Book:

Book Details

Book_Title

Author

Published Date

Page Numbers

ID	Book_Title	Author	Publish_date	Page_number
6	c++ programming	ali morad	2022-01-05 21:30:17.000	345
1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
4	Python	johanson mark	1900-01-01 00:00:00.000	600

1 reference

```
private void btn_update_Click(object sender, EventArgs e)
{
    try
    {
        cmd = new SqlCommand("Update Books_TB set Book_Title='" + texttitle.Text +
            "',Author='" + textauthor.Text + "',Publish_date='" + dateTimePicker1.Value +
            "',Page_number=" + textnumbers.Text + " where ID='" + txtshow.Text + "'", conn);
        conn.Open();
        cmd.ExecuteNonQuery();
    }
    catch (SqlException ex)
    {
        MessageBox.Show(ex.Message);
    }
    finally
    {
        reader.Close();
        conn.Close();
    }
}
```

The row of id=4 after update

ID	Book_Title	Author	Publish_date	Page_number
6	c++ programming	ali morad	2022-01-05 21:30:17.000	345
1	C++ basics	Jine Pual	2012-03-12 00:00:00.000	200
2	Programing in Java	Tomas Bear	2020-04-10 00:00:00.000	500
4	Mobile Programming	jasmon jack	2023-01-10 00:00:00.000	100

References

- Philip Conrod & Lou Tylee," Visual C #® And Databases A Computer Programming Tutorial", Kidware Software LLC.,2017
- <https://learn.microsoft.com/en-us/dotnet/api/system.data.sqlclient.sqlconnection?view=dotnet-plat-ext-8.0>
- <https://www.dotnetperls.com/sqlclient>